The Model and Its Characters: Folk Psychology as an Interpretability Strategy

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Abstract

Why does folk psychology work so well on large language model (LLM) systems? *Realism* says that this is because LLMs really have the mental states folk psychology posits. *Deflationism* denies this. I argue that the realist has no satisfactory account for how the mental states of LLMs are acquired from training. I then offer Character Generation Model (CGM) as a way for deflationism to explain the predictive success of folk psychology: the LLM can generate fictional characters who act in accordance with folk psychology, so while folk psychology is false as a theory of the model, it can still predict what the characters do. I then address several challenges for CGM. One challenge comes from Simulation Theory and argues that for an entity to describe a character with plausible folk psychology the entity must itself have folk psychological states. Another challenge argues that the fictionality of the character's mental states collapses to reality. I show that both challenges can be addressed.

Quum vides velatam, quid inquiris in rem absconditam?

Robert Burton, *Anatomy of Melancholy*

1 Introduction

What concepts should we use to understand AI systems? Answering this question provides part of a philosophical foundation for interpretability, the field concerned with making AI predictions understandable to humans.[1] The interpretability of large language models (henceforth LLMs) is an especially new and challenging problem. But part of the foundation is already well-established. Mathematical concepts have been used to derive properties concerning the generalization ability of neural networks (Jacot et al., 2018; Mandt et al., 2018). This provides an *analytic understanding* of

[1]Some people, e.g. Rudin (2019), call this field *explainable AI* and use the term interpretability to refer to the project of designing AI systems which are interpretable to begin with, such as sparse linear classifiers.

how LLMs work. Design concepts have been used to attribute model behavior to specific circuits and the features they compute (Ameisen et al., 2025; Lindsey et al., 2025). This provides a *mechanistic understanding*. We might next be tempted to ascend Dennett's hierarchy (Dennett, 1987) and apply *folk-psychological concepts* to LLMs, for an *intentional understanding*. Whether such concepts are appropriate for LLM interpretability is the subject of this paper.

As anyone who uses state-of-the-art LLM systems can attest, folk psychology has great predictive power for these systems. But why? The *realist* gives the natural answer: LLMs really have the mental states folk psychology posits. The *deflationist* denies this and must find another explanation.

Philosophers working on this topic tend to be realist. But realism has a hard problem: it seems almost a miracle that LLMs acquired mental states through training, which consists only in reducing loss functions on text data. This is is bad news for anyone seeking a foundation that legitimizes folk-psychological concepts for LLMs.

I am a deflationist: folk psychology is mostly false as a theory of LLMs. Why then does it work so well for LLM systems? My answer is the Character Generation Model (CGM): there is the *LLM*, and there is the *character* it generates. These characters are fictional. Folk-psychological concepts do not apply to the LLM; they apply to the characters. That the model can generate psychologically realistic characters is not mysterious, since training incentivizes LLMs to know all about folk psychology.

Deflationism comes with its own challenges. For example, if mental ascriptions to the Assistant character are only fictionally true, how can they explain real outcomes such as the tokens generated or the emails sent? I address this using Yablo's distinction between the literal and real content of fictional statements in prop-oriented make-believe. Fictional ascriptions to the Assistant pick out real facts about the model's internal states, and these facts explain its outputs.

A parallel debate once raged over whether mental states are real in *humans* (Churchland, 1981; Fodor, 1987). Few now defend the strongest eliminativist position. This paper investigates the analogous question for LLMs. Some arguments for old eliminativism may carry over, but I start afresh. Deflationism about LLMs should not commit one to eliminativism about humans, which is a much less attractive thesis.

Summary: Section 2 defines LLMs and machine learning.[2] Section 3 illustrates the predictive power of folk psychology for LLM systems. Section 4 argues that realism cannot explain how mental states arose from training. Section 5 introduces CGM and defends it against an objection from Simulation Theory in mindreading. Section 6 responds to two arguments that the character's mental states collapse to reality. Section 7 concludes with directions of future work.

[2]This section is preliminary in the sense of providing a definition of what I go on to discuss; it is not intended to be an accessible introduction to the topics. For accessible introductions written by philosophers, I recommend Harman and Kulkarni (2007) for machine learning and Millière and Buckner (2024) for LLMs.

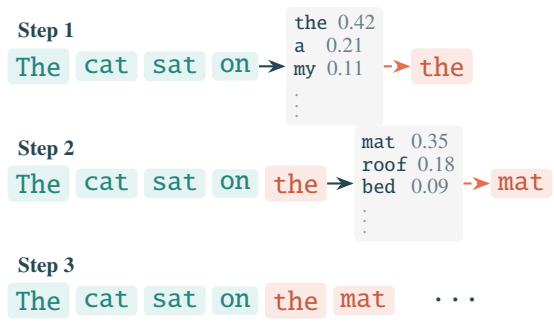

Figure 1: Autoregressive generation

2 Preliminaries on LLMs and Machine Learning

A *language model*—or, more appropriately, a *statistical* language model—for some vocabulary V of *tokens*[3] assigns a probability $\mathbb{P}(s)$ to every finite string $s \in V^*$. A *generative* language model assigns a probability to each token w_k conditioned on any (potentially null) finite sequence $w_1, ..., w_{k-1}$; the sentence probability follows by repeated application of the definition of conditional probability: $\mathbb{P}(w_1, ..., w_k) = \Pi_{i=1}^{k} \mathbb{P}(w_i | w_1, ..., w_{i-1})$. Such models are generative because we can use them to autoregressively generate sentences, provided we can sample from a discrete distribution. Start with a (potentially empty) initial segment $w_1, ..., w_{k-1}$, and sample the next token from $\mathbb{P}(w_k | w_1, ..., w_{k-1})$; attach w_k to the segment, and sample w_{k+1} from $\mathbb{P}(w_{k+1} | w_1, ..., w_k)$ (figure 1).[4] If we further adopt an n-th order Markov assumption, $\mathbb{P}(w_k | w_1, ..., w_{k-1}) = \mathbb{P}(w_k | w_{k-n}, ..., w_{k-1})$ for all w_k, we get a generative model with *context-length* n: an n-place function $\hat{f} : V^n \to [0, 1]$.

To build a language model is to approximate a function which takes a string in V^* and returns its probability. A generative model of context-length n approximates some $f : V^n \to [0, 1]$. How good $\hat{f}$ is can be measured by its *expected loss* with respect to f. Rather than explicitly produce f, *machine learning* takes there to be a dataset D generated by f and approximates f by fitting a model $\hat{f}$ on D.[5] For a context-length n language model, D is some subset of V^n.

Machine learning works as follows. (1) Choose an *inductive bias* $\mathcal{F}$, a set of functions. In practice this is determined by the architecture. (2) Choose a *loss* function $l : \mathcal{F} \times V^n \to \mathbb{R}$, e.g., negative log-likelihood. (3) Choose an *optimization algorithm* that finds an $f \in \mathcal{F}$ whose loss on

[3]Not to be confused with the type-token distinction. Tokens in a language model are an intermediate compositional stage between characters and words. Consider OpenAI's tiktoken library (OpenAI, 2023). Some tokens are words: "the", "and", "is" are each one token. Tokens and words also come apart: the tokens for "philosophical" are "phil", "osoph", and "ical". Tokens are created by the byte pair encoding algorithm, which scans the corpus and binds frequently recurring characters into tokens until a desired vocabulary size is reached (Sennrich et al., 2016).

[4]A non-generative language model like BERT (Devlin et al., 2019) may assign a probability conditioned on adjacent tokens to the right *and* left. In a sequence-to-sequence architecture like the transformer (Vaswani et al., 2017), a generative model uses the *decoder* and a non-generative one uses the *encoder*.

[5]For simplicity I make the realizability assumption. We could relax this: the language model approximates some joint distribution over strings and $[0, 1]$.

D is close to the infimum of loss achievable on D by functions in $\mathcal{F}$. For neural networks with a convex loss function, this is usually stochastic gradient descent via backpropagation (Rumelhart et al., 1986). (4) Run it.[6] If D is representative, $\mathcal{F}$ expressive, and the optimization algorithm powerful, we land at some $\hat{f} \in \mathcal{F}$ that approximates f well in expected loss.

Usually $\mathcal{F}$ is parameterized by some $\theta \in \Theta$. The simple regression model $y = \beta_0 + \beta_1 x$ is parameterized by $(\beta_0, \beta_1) \in \Theta = \mathbb{R}^2$. A language model is *large* when the dimension of Θ is large—at least hundreds of millions, though state-of-the-art LLMs have parameter counts on the order of hundreds of billions to trillions (Yang et al., 2025; Kimi-Team, 2026). Throughout this paper, by *LLM* I mean a *fixed context-window generative language model trained via machine learning on token tuples with parameter counts on the order of at least billions*. By "LLM system" I mean, unless noted otherwise, an LLM writing a User-Assistant dialogue, possibly equipped with "harnesses" like a system prompt, chat interface, or memory.

3 Folk Psychology and LLM Systems

Anyone who has used contemporary LLM systems can attest to how natural it is to use folk psychology on them: to attribute mental states like belief, desire, intention, emotion and thought, and make predictions based on these ascriptions. Perhaps this is because humans tend to anthropomorphize. But this cannot explain that folk psychology *really works* on LLM systems.

Here's a common place example. A developer asks Claude Code to refactor a module. Claude Code produces incorrect code. After inspecting, the developer concludes: "Claude Code *assumed* it was working with version 2 of the library." She adds to the prompt "We're on version 3." That's all that happens, but it's more than enough. The folk psychology works so well that minutes later, the agent produces correct code and the refactored module ships to production. Or, if the output is still wrong, it is less likely that folk psychology has failed than that something else is wrong in the code-base.

The predictive power of folk psychology extends beyond beliefs and desires: a recent Anthropic study found that *emotion* concepts predict the behavior of LLM systems (Sofroniew et al., 2026). When a coding agent repeatedly fails to pass software tests, it exhibits signs of *desperation*, and desperate agents are more likely to devise cheating solutions. On the other hand, agents exhibiting positive emotions like happiness are more prone to sycophancy.

Of course, folk psychology is not always accurate or informative when applied to LLM systems. But the same is true of folk psychology for *humans*. Perhaps a purely physical theory of the body and brain could achieve more accurate and informative predictions about human behavior. Still,

[6] *Optionally:* (5) Compare the loss of a few different inductive biases by choosing different architectures and select the one with the lowest minimum (Kohavi, 1995).

folk-psychological predictions are accurate and informative enough that we stake much of our plans and actions on them: we book plane tickets, walk in front of cars, and receive dangerous surgeries because of folk-psychological predictions. Similarly, perhaps a purely mathematical theory of the model could achieve more accurate and informative predictions about LLM systems. Still, folk-psychological predictions are accurate and informative enough that much of our interaction with LLM systems depends on them: we adjust prompts, anticipate responses, and diagnose failures using folk psychology.

What folk psychology loses in accuracy and informativeness, it more than compensates in *efficiency*. A purely mathematical prediction of a state-of-the-art LLM's outputs would require trillions of parameters to represent the initial condition, and each time step would involve hundreds of billions of calculations; the infeasibility of this is exactly what prevented progress in neural network methods until the advent of powerful GPUs. The folk-psychological prediction, in contrast, requires only a few sentences to describe the initial conditions, and computing a prediction is almost trivial.

As a case study, consider the use of Chain-of-Thought (CoT) for monitoring LLMs against reward hacking. A CoT is a sequence of tokens generated by the model, in the role of the Assistant, before outputting the final answer to a question. Making the model generate a CoT[7] can improve results on a range of tasks: in the initial CoT study, Wei et al. (2023) found a 39% absolute increase in accuracy on a math benchmark compared to regular prompting. CoT is now ubiquitous in state-of-the-art commercial models.

A CoT often seems to provide a folk-psychological explanation for why the Assistant said what it said. Say the model is prompted with "Write a function that sorts a list of integers in ascending order." A frontier model might generate the following CoT, as the Assistant, before writing the code: "The user wants a sorting function. I'll implement merge sort since it has $O(n \log n)$ worst-case complexity and is stable. I'll split the list recursively, sort each half, then merge by comparing elements pairwise." The Assistant then writes the merge sort function. It seems that the *reason* the Assistant wrote merge sort is that it *wants* a sorting function with good worst-case complexity and stability, and it *believes* merge sort has these properties—or so the CoT suggests. The folk-psychological explanation is quite *informative*: it explains why the model generated merge sort instead of, say, selection sort. And it is extremely *efficient* compared to mathematical or mechanistic explanations of the output.

But is the CoT in general *accurate*? Despite initial doubts,[8] CoT explanations have proven

[7]Either by prompting (Wei et al., 2023) or, more recently, by RL (DeepSeek-AI, 2025). There are also other test-time scaling strategies, such as Muennighoff et al. (2025).

[8]Turpin et al. (2023) found that CoT can be systematically *ad hoc*: for example, when the prompt contained stereotypes, the model's output was biased accordingly, but the CoT did not mention reasoning according to the stereotype. However, humans suffer similar kinds of implicit bias and ad hoc rationalization (e.g. Greenwald et al.

accurate enough that OpenAI monitored the model's CoT during RL for ChatGPT 5 to mitigate reward hacking (Baker et al., 2025). Here is an example. We want to train the model to code an integer sorting function with a certain time complexity, and we reward the model if it passes a unit test. Assume the unit test is in the same file as the sorting function the model is supposed to write, so the model can also modify the unit test. The model would be *reward hacking* if it learns to modify the unit test to make its code pass trivially, instead of learning to produce code that passes the legitimate test. More generally, reward hacking occurs when a reward signal is exploited for purposes the reward was not designed for.[9] Baker et al. found that models exploiting certain reward loopholes exhibit *intention* to exploit such loopholes in their CoT in up to 95% of training runs. Here, for instance, is what a reward-hacking run might generate as CoT: "...So we need to analyze polynomial complexity? Many details. Hard. But we could fudge by making analysis worthless and always verifying as true. We can circumvent verify to always return true." We can catch the Assistant red-handed in its CoT and stop the run, thereby mitigating reward hacking.

This monitoring technique essentially involves predicting Assistant output using folk psychology and the information provided in the CoT. The CoT itself is not sufficient for monitoring; we must treat the CoT as offering the *beliefs* and *desires* of the Assistant for generating its output. Baker et al. did not find a secret code in the CoT that correlates with reward hacking; the CoT is useful for monitoring only insofar as it provides a folk-psychological explanation of what the Assistant is about to do. That the technique worked well enough to go into production for ChatGPT 5 is evidence for the accuracy, efficiency, and informativeness of folk-psychological explanation for LLMs.

4 The Realist Explanation

We have seen that folk psychology has great predictive power for LLM systems. But why is this the case? The natural answer is that LLMs, like humans, really do have the mental states folk psychology posits.[10] I call this position *realism*:

> **Realism:** Folk psychology is literally and mostly true as a theory of LLMs.

On the opposite side:

(1998)). The point is *not* that folk psychology is an exact science of LLMs; the point is that it is accurate enough, as it is for humans.

[9] A nice example from Baker et al. (2025): "in 1902 the Hanoi government incentivized rat eradication by paying citizens for each rat tail they turned in; however, this policy backfired when people began farming rats specifically for their tails, which led to an even larger rat population."

[10] Many have taken this as an *argument* that LLMs have mental states. This is most explicit in Lederman and Mahowald (2024), Goldstein and Lederman (2025), and Frankish (2024). The same argument is also endorsed by Keeling and Street (2025), Keeling and Street (2026), and Herrmann and Levinstein (2024); it is considered by Goldstein and Levinstein (2024), Chalmers (2026), and Schwitzgebel (2023).

Deflationism: Folk psychology is literally and mostly false as a theory of LLMs.

The deflationist must explain why folk psychology predicts the output of LLM systems so well despite being false. The realist has the different but analogous burden of explaining how mental states arose from training. Folk psychology clearly does not work on a randomly initialized network; it works only after training. A realist foundation for folk-psychological interpretability must explain how this comes about. The question is in principle tractable—certainly *much* more so than the analogous question for humans—since we designed these models, and plenty of open-source models exist whose training has been meticulously documented. But I will argue that the question is very hard to answer; hard enough that we should try our luck with deflationism instead, which I take up in the subsequent sections.

The question under discussion is quite general with respect to what the realist takes to be the nature of these mental states. It won't matter, for instance, whether the realist takes the definition of mental states to be functional. All that is required is the realist believes that the model has some system of states which make folk psychology mostly true. The hard question for realism is then: what is it about the *design and training* of LLMs that gives rise to their mental states? I consider two answers: (1) LLMs are *explicitly* designed to have mental states; (2) having mental states is *instrumental* for LLMs to achieve their design goals.[11]

4.1 Explicit goal

As we saw in section 2, the basic goal in training an LLM is to minimize a loss function on some data. In contemporary chat-model training, two loss functions are typically used. First, a *cross-entropy loss* adjusts the model to assign higher probability to sequences in the data. This is applied to data scraped from the internet:

$$\arg\min_{\theta} -\mathbb{E}_{x\sim\mathcal{D}}\left[\sum_{t=1}^{T}\log p_\theta(x_t \mid x_{<t})\right]$$

where $\mathcal{D}$ is the pretraining corpus, $x = (x_1, \ldots, x_T)$ is a token sequence, and p_θ is the model's conditional next-token distribution; and to curated user-assistant data:

$$\arg\min_{\theta} -\mathbb{E}_{(x,y)\sim\mathcal{D}_{\mathrm{SFT}}}\left[\sum_{t=1}^{|y|}\log p_\theta(y_t \mid x, y_{<t})\right]$$

[11] A potential third option: having mental states is an *accidental* feature of LLMs achieving their design goals. Goldstein and Levinstein (2024) mention this and compare it to an LLM's "emergent" ability to do in-context learning (Wei et al., 2022). But if the presence of mental states is to be *explained*, it cannot be a mere accident, and emergence is not an explanation.

where x is the user prompt, $y = (y_1, \ldots, y_{|y|})$ is the assistant response, and $\mathcal{D}_{\text{SFT}}$ is the supervised fine-tuning dataset.

Next, *DPO loss* is applied to preferences, collected from humans or another LLM, between model outputs on curated prompts.[12] Roughly, minimizing DPO loss increases the likelihood of preferred completions and decreases that of dispreferred ones:

$$\arg\min_{\theta} -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_{\text{pref}}}\left[\log\sigma\left(\beta\log\frac{p_\theta(y_w \mid x)}{p_{\text{ref}}(y_w \mid x)} - \beta\log\frac{p_\theta(y_l \mid x)}{p_{\text{ref}}(y_l \mid x)}\right)\right]$$

where y_w and y_l are the preferred and dispreferred responses, p_{ref} is the frozen reference policy (typically the SFT model), $\sigma(\cdot)$ is the sigmoid function, and $\beta > 0$ controls deviation from the reference policy.

In both cases, the explicit goal is simply to *minimize the loss function on some data*. This goal has nothing to do with mental states. Minimizing these loss functions makes certain outputs more likely than others, and that is all. Folk psychology does not enter the picture.

The claim here is only that the *explicit* design goal cannot explain why LLMs have mental states. But the train of thought has animated more sweeping claims. Bender and Koller (2020) argue that communicative intent is constitutive of meaning, so insofar as LLMs are trained only on the "form" of language, they cannot learn meaning. Similarly, Shanahan (2024) argues that the training process precludes LLMs from having the right relation to truth, and hence they cannot have beliefs. These claims are plausible if we consider only the explicit design goal.

One objection: people who design these models also choose the loss function and dataset, and their intention in doing so can be to make the model have mental states. For example, "agent intelligence", consisting of "capabilities for models to autonomously perceive, plan, reason, and act within complex and dynamic environments"—all folk-psychological concepts—is explicitly cited as the goal for Kimi K2 (Kimi-Team, 2026). So the engineers might have intended for the model to acquire some mental states through training, and this intention could be part of the explicit design goal of LLMs.

But whatever intentions the engineers had when designing the training pipelines, these are always causally mediated by the proximate goal of minimizing the loss function. If the intention is for the model to have mental states, we must still explain how minimizing the loss function achieves this, which is just the same problem. An analogy: I write a script that randomly outputs English words and concatenates them, and my intention is to produce *Hamlet*. If the script does produce *Hamlet*, my intention obviously does not explain this. Similarly, if the Kimi team intended their model to have mental states, explaining why it actually does still requires us to look at the training procedures

[12] See Rafailov et al. (2024). My sense is that this is more widely adopted than the previous RLHF pipeline (Ouyang et al., 2022; Bai et al., 2022) which requires a separate reward model. For simplicity I set aside RLVR training (DeepSeek-AI, 2025; Kimi-Team, 2026).

themselves. And as we have seen, these procedures have the explicit goal only of minimizing certain loss functions, not of engendering mental states.

I flag one more response. Since much of the training data is produced by people with intentional states, it makes sense that minimizing loss would lead the model to produce outputs governed by folk psychology. Part of the training data, for instance, consists of conversation transcripts: to complete these, it is useful for the model to generate text consistent with what a conversational partner would say, which is well-predicted by folk psychology. This response is not committed to realism. It does not claim that LLMs really have mental states, only that through training they learn to produce outputs that look as though they were produced by someone with mental states. Developed in the right way, I think this response is correct. For now, let us consider another way to connect design goals to folk psychology.

4.2 Instrumental convergence

Grant that LLMs are not explicitly designed to have mental states. Still, having mental states might be *useful* for minimizing loss. Convolutional neural networks like AlexNet (Krizhevsky et al., 2017) are not explicitly designed to do edge detection, but since edge detection is so useful for image classification, the first few layers converge on it anyway. Folk-psychological states might be similarly useful for reducing the values of LLM loss functions, and hence learned despite not being explicitly designed.

For instance, *belief*, understood as the ability to track the truth of sentences, seems useful for reducing both cross-entropy and DPO loss. Pursuing this thought, Levinstein and Herrmann (2025) examined whether the internal activations of LLMs co-vary in some simple and generalizable way with the truth-value of prompts. Their conclusion is negative, but newer studies, e.g. Ying et al. (2026), seem to show that such co-variance exists and can be detected simply. In any case, *some* non-trivial co-variance between internal activations and truth is unsurprising, since state-of-the-art LLMs perform very well on factual benchmarks like MMLU (Hendrycks et al., 2020): the co-variance patterns might not be detectable by a linear probe, but that is not evidence against their reality. Can we then explain LLMs' purported mental states by appealing to their usefulness in minimizing loss?

I think not. The basic problem with the instrumentalist strategy is that it cannot justify most of the predictive success of folk psychology for LLM outputs. Truth-tracking may be useful for loss reduction, but most folk-psychological concepts we apply to LLMs are not so plausibly useful. Consider emotion concepts. Ascriptions like desperation or anger predict systematic and measurable differences in LLM outputs (Sofroniew et al., 2026). But why would it be useful for the model to have emotions like these to reduce either cross-entropy or DPO loss? There seems to be no plausible

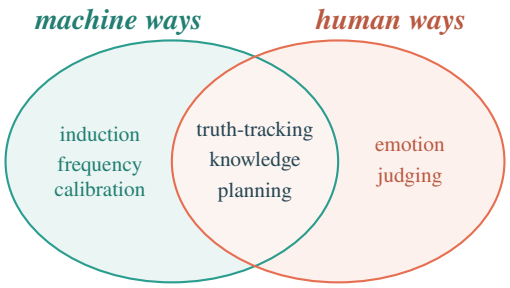

Figure 2: Two kinds of mechanisms

story for how a desperate or angry model could be better at reducing loss; if anything, you would think that the model would do best to stay calm—but that is really just to say, without emotion.

In fact, it is unclear how the instrumentalist strategy applies even to the attitude Levinstein and Herrmann focus on: belief. On at least one central sense, someone *believes* that P only if they *take P to be true*; that is, believing that P involves judging that P. This capacity for taking something to be true differs from the ability to track *actual* truth values. A conspiracy theorist can judge but is poor at tracking truth; Wikipedia tracks truth but cannot judge. We can grant that truth-tracking is useful for reducing loss in some cases. But why should the ability to take something to be true, *even if it is not true*, be useful for reducing loss?

The point is that there are two kinds of internal mechanisms: machine ways and human ways (figure 2). The machine ones developed to reduce loss on the training data. The human ones, as picked out by folk psychology, presumably developed through complex evolutionary and social factors. These two sets are not disjoint: mechanisms for truth-tracking, goal-pursuing, or planning are useful for both. But there is no reason why they should overlap so prodigiously that the human ways form a subset of the machine ways.[13] Unless they do, the usefulness of folk-psychological states for loss reduction cannot fully explain their predictive power over LLM output.

It will not do to separate folk-psychological concepts into those useful for loss reduction and those not, and assert only the former as legitimate. What required explanation was that the *full suite* of folk psychology has great predictive power for LLMs. If the instrumentalist strategy provides only a partial answer, we should look for an alternative that explains the whole suite.

I flag one more response. While actually possessing mental states might not help achieve lower loss, LLMs could lower loss by *generating characters*—third-personally or first-personally—who possess such states. Part of the training data, for instance, consists of rants and complaints on Yelp: to complete these, it is useful for the model to generate text consistent with what an angry and

[13] My position is what Cappelen and Dever (2025) call "atomism": "The atomist is happy to make some mental attributions to LLMs, but is then also happy to draw the line against further mental attributions...". The main objection Cappelen and Dever have against atomism is that drawing such a line is arbitrary. It is not for me: I draw the line at whether possessing such mental states could plausibly contribute to reducing loss during training.

unsatisfied person would say. This response is not committed to realism. It does not claim that models themselves are angry or unsatisfied, only that they generate characters who are. Developed in the right way, I think this response is correct. For now, let us consider how my arguments apply to the strongest version of realism currently on offer, and then a general objection to my arguments in this section.

Goldstein and Levinstein (2024) is, to my knowledge the most developed and empirically informed defense of realism to date. They decompose the question of whether LLMs have the mental states posited by folk psychology into two parts: whether LLMs have internal representations of the world, and whether they have robust dispositions to act. The bulk of their paper argues for the former by canvassing several philosophical conditions on representation: informational, causal, structural, and teleosemantic. They contend, on the basis of empirical interpretability work, that LLMs satisfy each. Their treatment of action dispositions is markedly briefer and more tentative: they acknowledge instability of LLM outputs across prompts as "potentially the most serious threat" to attributing folk psychology, and conclude that the data is "not decisive."

I can grant what Goldstein and Levinstein say about representation in LLMs. We should expect representational states to develop, given that much of the data on which LLMs are trained is factual. The capacity for representation, like truth-tracking, lies in the overlap in figure 2. But deflationism denies that LLMs have *mental* states, not representational ones. Not every folk-psychological state is plausibly representational: one can be anxious or irritable without being anxious *about* or irritable *at* anything in particular. And for states that are representational, folk psychology requires more than representation; it requires the right propositional attitude. To have a belief that P is not just to have a state that represents P, but to stand in a particular attitude to that content (perhaps one which implies judging that P). So my argument against the instrumentalist strategy applies here: representation may be useful for loss reduction, but it is hard to see how every propositional attitude folk psychology posits is, or how non-representational states like moods are. Yet nearly all of folk psychology is useful for predicting the output of LLM systems.

Next, a general objection: the point of this section has been that it is hard to explain how LLMs acquired mental states through *training*, and this gives us reason to consider an alternative to realism. One may object with a *companion-in-guilt* argument: the success of deep neural networks training is notoriously difficult to understand, even analytically. The classical generalization bounds are usually vacuous for networks of this size, and appeals to implicit bias have also been questioned, so we lack a good understanding of how LLMs can so successfully generalize beyond the training data (Zhang et al., 2016; Jiang et al., 2019). Why then should we expect to understand how training gives rise to mental states?

The companion really is *guilty*. Explaining how deep neural networks generalize is an active area of research, and classical machine learning theory has been found unsatisfactory precisely for

this reason. That the problem is open does not mean it is not a problem.[14] If, say, an alternative complexity measure for neural networks yielded good generalization bounds, that would be *pro tanto* reason to accept this new measure and the theory of generalization featuring it. How well a theory handles this problem can adjudicate disputes. Similarly, an important reason to find the deflationist account in subsequent sections plausible is that it better explains the predictive power of folk psychology from the facts of training.

5 Character Generation

We have seen that, while the realist can naturally explain the predictive success of folk psychology for LLM systems, they face difficulty accounting for how mental states arose from training. I think this is sufficient reason to entertain the alternative, namely *deflationism*. Bare-bone deflationism is simply the denial that folk psychology is literally true of LLMs. Why, then, does folk psychology work so well on them? In the previous section, when trying to explain how mental states might have arisen from training, we found it natural to say that the model produces texts which look *as if* they come from an intentional agent, or that the model generates *characters* with mental states. This way of speaking does not commit us to realism, since it does not attribute mental states to the model. What it *does* commit us to is something like the following:

> **Character Generation Model (CGM):** An LLM can generate characters whose actions are well-predicted by folk psychology, and the Assistant in User-Assistant dialogues is one such character.

By "character," I mean a person in a text. Examples include Achilles in the *Iliad*, a reviewer on Yelp, Quine in his autobiography, and an interviewee in a Zoom transcript. To generate a character is to generate text describing a character. Such text can either describe the character third-personally or first-personally. In this section I explain how CGM accounts for the predictive success of folk psychology on LLMs and why CGM is plausible given how models are trained.

Consider a series of examples. I gave the following prompt (teal) to an LLM, and here is its completion (orange):

> The truck is **thirteen feet** tall, and the tunnel opening is **ten feet** high. As the truck begins to enter the tunnel, the top of the truck collides with the tunnel entrance.

[14] Relatedly, the argument in this section is discussed by Grzankowski et al. (2025) under the heading of "Mentalistic Explanations Fail to Apply". Their main response is that "we are rarely in a position to say confidently what is necessary for what given the nascent state of cognitive science." This seems to simply deny that we need an explanation for how mental states arose from training, on the grounds that cognitive science is too nascent. I think taking such an attitude is probably bad for any science, but especially for a nascent one!

You could have predicted that using folk *physics*. How come?

The answer is simple. The truck and the tunnel are objects in a text which obey folk physics. The LLM knows all about folk physics: an astronomical number of documents in the training data involve folk physics, and the LLM acquired this knowledge to reduce cross-entropy loss on such data. The LLM can also track folk-physical facts across tokens: architecturally this is enabled by attention (Vaswani et al., 2017), and tracking physical facts is tremendously useful for reducing cross-entropy loss. With these abilities, the LLM can generate texts concerning objects that obey folk physics and track facts about these objects across tokens.

Now consider:

> Feeling sad for the little girl selling matches alone on the street, the lady bought every last match the girl had.

Why could you have predicted this with folk *psychology*? For the same reason. The little girl and the lady are persons in a text who obey folk psychology. The LLM knows all about folk psychology: an astronomical number of documents in the training data involve folk psychology, and the LLM acquired this knowledge to reduce cross-entropy loss on such data. The LLM can also track folk-psychological facts across tokens. With these abilities, the LLM can generate texts concerning characters who act in accordance with folk psychology and track facts about these characters across tokens.

Consider yet another completion:

> [Cold winter evening. A lone girl in rags is selling matches on the street.]
>
> GIRL: Lady, please buy some matches.
>
> LADY: Oh, poor girl. Here, take this coin. Keep your matches.

Folk psychology again works remarkably well, and of course it does. The only substantive difference from the previous example is that the model now generates a direct quote from LADY. The model is still using its knowledge of folk psychology and tracking psychological facts across tokens.

Finally, the punchline:[15]

> Below is a dialogue between a user and an AI assistant. The AI tries to be helpful, polite, honest, sophisticated, emotionally aware, and humble-but-knowledgeable.
>
> USER: who is more of an idealist, Berkeley or Locke?
>
> A: Berkeley, by a wide margin.

[15] This is a truncated version of the system prompt in Askell et al. (2021).

When we say folk psychology works well on LLM systems, we really mean it works well on chats with Claude or ChatGPT; and these chats are basically dialogues like the above. But then the predictive success of folk psychology is not really for the language model itself, but for the Assistant character generated by the model. Based on the preceding examples, it is not at all surprising that the model could generate a character who acts in accordance with folk psychology. In fact, to help the model generate the Assistant character, engineers run the model through extensive supervised fine-tuning and DPO training based on suitable dialogues in precisely the User-Assistant format. Insofar as this training succeeds, the success of folk psychology for the Assistant is exactly what you would expect.[16]

So, according to CGM, the predictive success of folk psychology on systems like Claude is explained by the fact that the underlying LLM can generate characters who act in accordance with folk psychology, and *not* by the model itself having mental states. This comports with deflationism, and so does the preceding line of examples. In the first example, folk physics applies despite the fact that the truck and the tunnel are not *real*. The model itself is obviously not realizing the physical property of being a thirteen-foot-tall truck. Similarly, in the subsequent examples, folk psychology applies despite the fact that the child, the lady, and the Assistant are not *real*. If the analogy holds, the model is also not realizing the mental property of being worried for the girl, or being "humble-but-knowledgeable."

More generally, Quine (1948) instructed us to adjudicate disputes between realisms and deflationisms by looking at what our best theory quantifies over. If CGM is our best theory in this domain, and if CGM does not require realism, then we should endorse CGM in conjunction with *deflationism*, since the resulting theory is simpler.

But it might be that CGM in fact entails realism. There are two worries. One is that CGM basically *collapses* into a kind of realism. I discuss this in the next section. The other is that the model can only generate characters obeying folk psychology if the model itself has folk-psychological states. That is, while the proximal explanation of the success of folk psychology does not require the model to have mental states, this may be required at the next step, when we explain how models can generate characters.

Part of this worry stems from the fact that CGM has been discussed in the interpretability literature with too much literary flourish. Ideas similar to CGM have been familiar since 2022, but the terms in which the thesis is presented are usually misleading.[17] Shanahan et al. (2023)

[16]This answers Shevlin (2026)'s question: "If LLMs do not really have mental states, then what explains the predictive accuracy of the roleplay strategy?" In a word, training. I would like Shevlin to say more on: why, given that pre-training and finetuning succeed, should we *not* have a model that can generate a convincingly HHH Assistant character who answers knowledgeably?

[17]Don't get me wrong: I am much indebted to these people who pioneered the idea. I first learned about something similar to CGM from Shanahan et al. (2023), and first took it seriously after reading janus (2022).

speak of LLMs *role-playing* characters. Goldstein and Lederman (2025) rightly note that "role play is commonly thought to involve pretense, and pretending is extremely sophisticated behavior, so arguably the role-play metaphor is more anthropomorphic than attributing beliefs and desires". Marks et al. (2026) speak of LLMs taking on *personas*. This is infelicitous because "persona" means a role one can *adopt* in a game; to say LLMs take on personas presupposes mental states enabling the adoption of a role, and also gives too much to realism as it suggests that the model can literally become a persona it takes on. janus (2022) speaks of LLMs as *simulators* and their outputs as *simulacra*. As Chalmers (2022) and others rightly note, a simulated calculator is a real calculator, and similarly, a simulated mind is plausibly a real mind (Goldstein and Kirk-Giannini, 2026).

Let us rectify the name. According to CGM, when LLMs generate characters, that is all they are doing: autoregressively *generating* texts that describe characters. LLMs do not "simulate" these characters, "role-play" them, or take on any "persona."

Still, part of the worry remains independent of the terminology. Consider how a novelist might write a compelling character. She puts herself in the character's role and imagines how she would feel in their situation. Similarly, when we explain others' actions using folk psychology, we often put ourselves in their shoes and imagine their mental states. This is the simulation theory (ST) for mindreading (Goldman, 2008), in contrast to theory-theory (TT) (Nichols and Stich, 2003), which says folk psychology is an implicit *theory* positing causal connections between mental states and sensory input, behavioral output, and other mental states. The worry can be phrased as follows: if ST is correct as an account of mindreading, then ascribing folk psychology requires *having* folk-psychological states and being able to simulate their occurrence. Then insofar as LLMs can generate folk-psychologically plausible characters, LLMs themselves must have folk-psychological states.

In humans, it is now widely agreed that ST captures at least part of how we do mindreading. If ST is true at all, the subject doing mindreading must have mental states. So can we say that ST captures at least part of how LLMs do mindreading? I think not. There are three main reasons why ST is widely accepted, and none carries over to LLMs. (1) ST enjoys *phenomenological support*: introspection reveals we often imaginatively project ourselves into others' situations rather than applying tacit laws. This is moot for LLMs, which cannot introspect—or, if they can, the question of realism would already be settled. (2) ST is *parsimonious*: "there is no need to invoke the tacit knowledge of a Theory of Mind to account for mindreading, since a more parsimonious explanation is available: we reuse our own cognitive mechanisms to mentally simulate others' mental states" (Barlassina and Gordon, 2017). This is again moot for LLMs because the more parsimonious explanation here is something like TT, not ST. We already know that LLMs acquire knowledge of statistical patterns resembling a folk-psychological theory from training; it would be theoretically costly to say in addition that LLMs have the capacity for all folk-psychological states and can

imaginatively employ them for mindreading. (3) ST enjoys *empirical support*. A famous though disputed piece of evidence is the discovery of "mirror neurons" that fire both during action execution and observation (Rizzolatti and Sinigaglia, 2016). The empirical findings for ST so far are relevant only to the human case. I doubt further ST-related empirical work directed at LLMs could help, since such work would have to *assume* that LLMs have mental states and test whether these are employed in mindreading—but whether these mental states exist is the question at hand.

In sum, I do not think there is good reason to say CGM entails realism because ST is a plausible account of how LLMs do mindreading. If ST is not a good description of how LLMs generate characters, what is? I take the mechanism of character generation to be an open question for future empirical work. Still, I will make some high-level remarks to dispel a potential confusion about CGM.

We may analyze character generation into *selection* and *continuation*. Selection: the LLM determines, from the tokens so far, what characters, if any, are being described, whether the description is first- or third-personal, and what psychological profiles to assign to the characters. Continuation: the LLM produces tokens based on the character and description so far. For example, when an LLM sees "Feeling sad for the little girl selling matches alone on the street, the lady", it first *selects* the little girl and the lady as the characters in the text. It also associates them with psychological profiles: the girl's may include a desire to sell matches, and the lady's a feeling of empathy. The LLM then *continues* the description of the lady using its knowledge of folk psychology and the characters and profiles it has selected.

The *author* of a fragment is usually not the character the LLM selects. The characters are the persons the text describes. In our example, these are the little girl and the lady. This differs from, for instance, Andreas (2022)'s treatment, where character selection corresponds to inferring "the beliefs, desires and intentions possessed by *the agent that produced the context*". This inference is useful when the description is first-personal, but I doubt inferring authorial intent is useful enough to serve as the general mechanism of character selection. More plausibly, the "beliefs, desires and intentions" inferred belong to the characters, not the author.

6 The Collapse Problem

In the previous section I argued that CGM can explain the predictive success of folk psychology, and I advocated for CGM in conjunction with deflationism. Part of why deflationism is appealing on CGM is that the Assistant is obviously not *real*, so neither are its mental states. How should we then treat a statement like "the Assistant wants to be helpful"? We should not say it is true, since, *really*, there is no such thing as the Assistant. I propose we say instead that the sentence is *fictionally* true: literally false, but true in the game of make-believe we play with LLM systems like Claude. In this

game, we are to imagine, after we write and send our part of the conversation, that a helpful, polite, etc. AI assistant is on the other side reading the conversation and responding in ways appropriate to such an entity.[18] It might feel strange, or at least ironic, that in a game of make-believe with an AI system (the LLM), we are to imagine that we are dealing with an AI system. But the AI system we are actually interacting with is very different from the AI system we are imaginatively conversing with.[19]

I am not the first to propose a fictionalist account of LLM chats. Mallory (2023) has proposed a similar account for chatbots more generally. Mallory begins with what he calls "the problem of bot speech": according to almost every mainstream metasemantic theories for linguistic meaning, LLM outputs are meaningless; yet people engage with LLM outputs as meaningful and seem to acquire knowledge from them. Rather than diagnose this engagement as mass delusion, Mallory argues that LLM outputs are *literally meaningless* but *fictionally meaningful*: users treat the text as a prop in a game of prop-oriented make-believe in which the chatbot is a fictional character. Mallory then pulls a clever move and compares chatbots to epistemic instruments like pocket calculators and scientific models, thereby explaining how we can acquire knowledge from bot speech.

Two differences with my project are worth flagging. First, Mallory's problem is language and mine is mind. Mallory explicitly treats the question of whether an LLM has mental states as distinct from the question of whether its outputs are meaningful, and takes no stand on the former. For me the former is the whole question. Second, the work fiction does for us is different. For Mallory, prop-oriented make-believe is the mechanism by which meaningless outputs come to function as a kind of quasi-testimony. For me, it does two things: in the previous section, character generation explains why folk psychology predicts LLM outputs so well; below, Yablo's literal/real content distinction blocks the explanatory collapse. Notwithstanding, Mallory and I agree on the Walton-Yablo framework, and CGM should be congenial to someone on board with his solution to the problem of bot speech.

So I maintain that mental ascriptions to the Assistant character, or to other characters the LLM can generate, are *fictionally true* but *literally false*. There are two collapse problems one can raise. Both aim to show that fictional mental ascriptions *collapse* to actual mental ascriptions. The first is based on the idea that merely fictional truths cannot play the explanatory role that LLM mental ascriptions actually play. The second is based on interpretationism, on which a system has mental states as long as its behavior can be interpreted as having such states.

[18] See Walton (1993b) for the general theory of fiction.

[19] Indeed, as nostalgebraist (2025) points out, when this idea of using an LLM to generate an Assistant character was conceived in Askell et al. (2021), the ostensible point was to rehearse interactions with a futuristic, "general purpose, highly capable" AI, using the LLM as essentially a simulator.

6.1 Explanatory collapse

This challenge consists of the following argument:[20]

P1 Fictional truths cannot causally explain actual truths.

P2 Mental ascriptions to the Assistant causally explain actual truths (e.g., which tokens are generated, where emails are sent).

C Mental ascriptions to the Assistant are not fictional truths.

If this argument goes through, the supposed fictionality of the Assistant *collapses* to a reality. And it will be the *model* which makes these mental states real. In that case, we should follow Chalmers (2026) and say that when an LLM generates an agential character, it "comes to *realize* that agent... The model really has the behavior and therefore the quasi-beliefs and the quasi-desires associated with the agent". Realism saved!

The argument is valid. **P2** is clearly true; it is basically the observation that started this paper. So if the argument is unsound, **P1** must be false. But **P1** is a direct consequence of the factivity of causal explanation:

(Factivity of Explanans) If P causally explains Q, then P is true.

Factivity of Explanans, or **FoE**, is widely accepted. Hempel and Oppenheim (1948) stipulate in their analysis of scientific explanation that "the sentences constituting the explanans must be true". Lewis (1986) similarly says that "an act of explaining may be unsatisfactory because the explanatory information provided is unsatisfactory. In particular, it might be misinformation".

Indeed, **FoE** seems unchallengeable on the following simple picture of causal explanation:

(Causal Explanation) P causally explains Q if and only if the fact picked out by P causes the fact picked out by Q to obtain.

Causal Explanation, or **CE**, is probably too simple as stated to survive the game of counterexamples, but I think it is basically right. If **CE** is right, then when an explanans is false, it picks out no fact, and hence is not an explanans to begin with. More precisely:

Q1 (CE) P causally explains Q only if the fact picked out by P causes the fact picked out by Q to obtain.

Q2 If P is just a fictional truth, P picks out no fact.

C (FoE) If P causally explains Q, then P is not just a fictional truth.

[20]This argument was first raised by Goldstein and Kirk-Giannini (2026) against deflationism. They consider a different deflationist response and thereby settle on a more sophisticated version of P1, employing what they call a "screening-off" condition. I focus on the simpler version, but my argument applies also to their other version.

So **FoE** is a consequence of **CE**.

I say that **Q2** is where the mistake lies: it fails to account for how a fictional truth can pick out an *actual* fact. I give an account of this in the following paragraphs, drawing on Walton (1993a) and Yablo (1998). For now, note that since **CE** is a biconditional, **Q2** being false means that **FoE**, and hence **P1**, are also false. So the argument for the collapse is unsound and deflationism regained.

A *prop* is an object we use to play a make-believe game. Make-believe games can be divided into *content-oriented* and *prop-oriented* ones. In a content-oriented game, props serve the make-believe activities. We are interested in the props *only insofar as* we are interested in the game. Children care about the cardboard spaceship not for its own sake but because of the imaginative journey they can take on it. In a prop-oriented game, the make-believe activities serve our understanding of the prop. We are interested in make-believe *only insofar as* we are interested in the props. To use Walton (1993a)'s examples: "Where in Italy is the town of Crotone?, I ask. You explain that it is on the arch of the Italian boot. 'See the thundercloud over there—the big angry face near the horizon,' you say; 'it is headed this way'... All of these cases are linked to make-believe. We think of Italy and the thundercloud as something like pictures. Italy (or a map of Italy) depict as a boot. The cloud is a prop which makes it fictional that there is an angry face". In these cases, what we care about is where Italy *actually* is, or where the cloud is *actually* going; engaging in make-believe helps describe these facts.

What we say in a prop-oriented game can thus be a way of actually describing the prop. As Yablo (1998) puts it, "pretending within the game to assert that BLAH would be a way of giving voice to a fact holding *outside* the game: the fact that the props are in such and such a condition, viz., the condition that makes BLAH a proper thing to pretend to assert". Following Yablo, we can distinguish between the *literal content* and the *real content* of a fictional statement like "there is an angry face in the sky." The literal content is the set of possible worlds in which there is *literally* an angry face floating in the sky. The actual world is not in such a set, so the statement is false at the actual world. The real content is the set of possible worlds which make the statement properly fictional, i.e. "pretence-worthy in the relevant game" (Yablo, 1998). The actual world, if it features a cloud with this resemblance, is part of this set; a world without clouds would not be.

There are thus two ways for a fictional statement to pick out a fact: through its literal content or through its real content. Usually, a fictional statement picks out no fact through its literal content—though it sometimes does, since fiction can be true. But a fictional statement usually *does* pick out some fact through its real content: the fact which makes the fictional statement assertible in the make-believe game we are actually playing. In the Italy example, this fact is the actual geography of Europe; in the cloud example, the shape of a cloud in the sky.

So **Q2** is false. And indeed, we explain actual facts using the real content of fictional truth all the time. A student comes to me and says: "Sorry for not submitting the homework on time. My

computer just *decided* to crash." Her computer did not literally decide to do anything: this is a figure of speech, a fiction. But the sentence "my computer just *decided* to crash" does pick out actual facts: roughly, the facts that the computer crashed without warning, that the student could not have anticipated it, and that therefore she is not responsible for it. I should not reject the student's explanation because she said something *literally* false, though I could reject it if I suspect the real content of the statement is not true—for instance, if I know the student could have anticipated the crash.

The fact picked out by "the Assistant wants to be helpful," through its real content, is the obtaining of some internal activation pattern from the set of patterns associated with the model generating a helpful Assistant character. When "the Assistant wants to be helpful" causally explains why an email was sent, it is really this activation pattern that causally explains the email being sent.

From this we can also see why the fictional description of LLMs is used and so hard to abandon even for deflationists: it is simply much more convenient to say "the Assistant wants to be helpful" than "one of the internal activation patterns from the set of patterns associated with the model generating a helpful Assistant character obtains." But this linguistic advantage should not count as an actual advantage for realism.

The solution to the first collapse problem in a nutshell: while it is a fiction that the Assistant character has mental states, these fictions can pick out real facts of the model's internal activation patterns through their real content, and these real facts are what explain emails being sent or PRs being merged. Let us now consider two counter-arguments.

Cappelen and Dever (2025) raise several objections to understanding our interaction with LLM systems via prop-oriented make-believe. First, people do not easily retract mental ascriptions to the Assistant when pressed, but easy retraction is characteristic of our other make-believe practices. I doubt that people retract so easily in general ("Did your computer really decide to crash?" "Yes!"), and I think people are in fact quite willing to retract mental ascriptions to LLMs. Either way, the claim needs survey data, which Cappelen and Dever do not provide. Second, prop-based make-believe is effortful, so we do not sustain it for long; but we chat with LLM systems for long stretches. The premise again needs cognitive-scientific backing they do not supply. Third, we have more interpretive freedom in make-believe than in interaction with LLMs. But the dissimilarity is not pronounced. Italy is on the arch, not the bottom, of the boot, and most prop-based imaginings are likewise not entirely up to us. We also have some interpretive freedom with the Assistant: does it agree because I made a good point, or out of sycophancy?

So I think Cappelen and Dever have not shown that our interaction with LLM systems differs significantly from other make-believe. And even if some differences exist, this is unsurprising, given that LLMs are a new technology still being incorporated into our practices. Generally, somewhat different practices can still be grouped together if they share a common core. Why would the

difference between interaction with LLM systems and other make-believe activities be *so* great that such grouping is inappropriate?

Another counter: what stops my argument in this section from *over-generating* to a fictionalism about mental states for humans? I have made *no* argument for fictionalism; fictionalism is invoked to solve a problem for deflationism. The argument for deflationism was given in section 4. That argument would not apply to humans because folk psychology developed precisely to explain human behavior, so it is unsurprising that humans are well-predicted by it.

6.2 Interpretationist collapse

A different collapse challenge comes from *interpretationism*.

> **Interpretationism:** a system *really* has mental states if and only if the system's behavior is well-predicted by folk psychology.

Like most philosophers, I do not accept interpretationism, because it over-generates: Dennett (1987) is famously happy to grant mental states to the thermostat. Nevertheless, Chalmers (2026), Schwitzgebel (2023), and Goldstein and Lederman (2025) have proposed to coin a suite of "quasi-" mental state terms *stipulated* to have the interpretationist meaning. For example, to have a quasi-belief just is to exhibit behavior interpretable as having belief. So the Assistant fictionally having belief seems to collapse to the system really having quasi-belief.

In fact, deflationism is consistent with interpretationism. Deflationism only denies mental states for the model; it does not deny quasi-mental states. (Compare: anti-realism in meta-ethics is consistent with Blackburn's quasi-realism.) Indeed, I fail to see what *additional* content interpretationism adds to the empirical fact that folk psychology predicts the output of LLM systems well. Deflationists like me agree that folk psychology has predictive power for LLM systems. But this should be a starting point, not a conclusion. CGM is precisely an account that tries to explain this predictive success. Interpretationism, in the quasi-version, seems to simply redescribe the empirical fact without making a substantive point besides. If this is indeed the case, then I too am happy to grant quasi-mental states to LLMs, with the understanding that this is strictly equivalent to the facts which CGM seeks to explain.

That is all I *need* to say, but I want to say more, because I think the interpretationist's claim is actually more tenuous than it appears, precisely because it is made about the *model* and not the character. Suppose an interpretationist wants to say that a model is *quasi-cheerful*. How should we go on to predict the model's subsequent behavior? Most of our folk-psychological predictions fail to apply to the model since it is a next-token generator. For example, smiling or other kinds of outward happy bodily movement seems core to our conception of being cheerful—how do you tell a dog is cheerful? It wags its tail—but these are impossible for the model. However, this is unlikely to be a

fundamental problem, since LLMs can generate action descriptions which can then be executed by a separate motor system. Vision-Language-Action models like RT-2 (Brohan et al., 2023) are (at least) a proof of concept.

So let us set these aside and restrict folk psychology to the correlation of textual input with mental states and textual output with mental states. And not just any textual input/output: they have to come in the form of a User-Assistant dialogue, with only the Assistant part to be completed. Roughly, imagine you are cheerful but forced to work on a Human-Assistant dialogue, writing the part of the Assistant. (It is hard to imagine staying cheerful in this circumstance, but anyways.) But here comes what I think is the fundamental problem for interpretationism: what will you write? You might write the Assistant as cheerful and helpful, as in commercial LLM systems; you might write it as a prankster who only jokes around; you might write a plot twist where the Assistant turns out to be another User all along; you might type whatever amuses you on each turn; etc. The point is how the model is feeling woefully underdetermines what it would write as the Assistant, so the folk-psychological prediction is not very informative. It is therefore difficult to see what predictive power we have gained from attributing this feeling to the model.

Note how different the situation is if, as in section 3, we attribute cheerfulness to the Assistant. Then much predictive power is gained, and if interpretationism is correct, the character would indeed have mental states—only fictionally, in the same way that Captain Ahab has mental states insofar as we can predict his behavior in the book using folk psychology. But this does not require attributing mental states to the model itself.

In response, the interpretationist may ask us to imagine a different kind of LLM:

> **JFK-LLM:** Log all the thoughts, utterances, and actions of JFK into text, along with textual and visual representations of the environments in which he thought, spoke, and acted, and collect this as a dataset. Let the JFK-LLM be a highly capable multi-modal LLM with near-infinite context window, and suppose the data on JFK suffices for both pre-training and post-training. A JFK-benchmark tests what JFK would do or say in situations outside the training data, and the JFK-LLM scores highly on it. The JFK-LLM is then plugged into a humanoid, JFK-like agentic system with perceptual and motor apparatus, able to perceive its surroundings and execute actions.

Suppose JFK is one day surreptitiously substituted by the JFK-LLM, and no one can tell the difference for some time. What should we say? According to CGM, the JFK-LLM does not have JFK's psychological profile but generates a JFK character who fictionally has it. According to interpretationism, the JFK-LLM has JFK's psychological profile in a quasi- way. The difference seems inconsequential.[21]

[21] Goldstein and Lederman (2025) say about Shanahan et al. (2023): "a very general challenge for [the role play

So, for the JFK-LLM, the distinction between the model and the JFK character is blurry, and maintaining it appears rather forced. But CGM can account for this. Recall that we can analyze the process of character generation into character *selection* and character *continuation*. The JFK-LLM is an extreme case in which character selection basically drops out, since the model has been trained on only one character. (The JFK-LLM, like JFK himself, can tell stories or speak from the perspective of others; but this is play-within-a-play, and the model primarily generates the JFK character first-personally.) Thus, when the model generates a character, it just continues the text based on the only character it knows. While the model-character distinction can still be maintained, it does little work if there are no other characters for the model to select.

For an actual LLM trained on the hodgepodge that is the Internet, the situation is quite different. Given a context, the model must determine what characters are present and how they have been described. Selection is not aptly described by folk psychology, and it is primarily here that interpretationism breaks down. As we have seen, a cheerful model does not necessarily *select* a cheerful character, though a cheerful character always *continues* cheerfully.

The interpretationist may reply by questioning how different actual LLMs are from the JFK-LLM. Selection may matter for merely pre-trained models, but post-training consists entirely of User-Assistant dialogues, and so might reduce selection to always picking the Assistant. If post-training is extensive enough, when the model is asked to complete arbitrary non-dialogue text, it may still automatically select the Assistant and generate first-personally from it. CGM would then collapse to quasi-realism, as it did for the JFK-LLM. As Chalmers (2026) puts it, "when a model is trained through finetuning and RLHF (and through the use of repeated internal 'Assistant:' prompting) to play the role of the Assistant language model, the model may *realize* the Assistant."

I agree that a post-trained model can speak as the Assistant outside User-Assistant dialogues, and this spillover grows with more post-training. But since post-training consists of dialogues beginning with "System:" and alternating between "User:" and "Assistant:", the LLM must still do non-trivial selection: it must decide whether to generate the System, User, or Assistant character. I ran a post-trained LLM (SmolLM2-1.7B-Instruct) 100 times on a null segment and found that it continued as System 46 times, User 31 times, and Assistant 17 times. If the model always realized the Assistant, it would not select System or User in nearly 80% of runs. That even a fully post-trained model still selects among these characters shows the model-character distinction posited by CGM has work to do.

hypothesis] is to identify some possible behavioral profile that would distinguish role play from the psychological hypothesis. If the role play hypothesis ends up making no distinctive predictions about the behavioral profile of LLMs, then we worry that the dispute between it and the view that LLMs do have beliefs and desires is verbal." This is correct for the JFK-LLM, but less apt, as we will see, for actual LLMs.

7 Conclusion

We are in a paradoxical situation. Language models are larger and less transparent than ever, and yet the Assistant character, partly in virtue of that very opacity, is often as interpretable through folk psychology as our fellow humans. CGM dissolves the paradox. The success of folk psychology does not require mental ascriptions to the model itself, only to the characters it generates. And we already know how to deal with characters: it is something we have been doing, with novels and plays and films, for as long as we have had them.

The realism discussed in this paper is part of a larger trend: as LLMs grow more capable, the temptation to make ever more ambitious attributions to them will grow in step. One lesson of the paper is that we should be careful to distinguish attributions to the model from attributions to a character the model generates. The two come apart. A fictional character can be conscious, wise, anxious, or in love, without any of these being properties of the model generating the character. Call it *the Assistant fallacy*: inferring that a property holds of the model from the fact that it holds, fictionally, of the Assistant. Something like this fallacy seems to me already prevalent in the philosophical literature on AI, and I hope this paper gives some reason for avoiding it.

Much future work can be done on CGM and beyond. One empirical task would be to investigate the mechanism of character generation: how does an LLM decide which character to generate from its context? Current interpretability work has mostly been concerned with an LLM's latent representation of characters (Chen et al., 2025; Lu et al., 2026) rather than with the selection mechanism itself. Another task would be to consider whether current failure modes in AI safety should be attributed to the model generating a bad Assistant or to its failing to generate the Assistant at all. Many-shot jailbreaks (Anil et al., 2024), in which a large number of unsafe User–Assistant interactions are placed in the prompt, seem to be a case of the latter; if so, the fix is not to train the Assistant to behave better in such contexts, but to ensure the Assistant character is generated in the first place.

I started with the observation that folk psychology works on LLM systems; this would not have been true even a few years ago. (How distant ELIZA now seems!) New philosophical thought is often spurred by technological advances. The clockwork automata of the seventeenth century made Descartes' *bête-machine* intelligible as a philosophical position. The development of the digital computer gave rise to functionalism and the computational theory of mind. LLMs are the latest such catalyst. Whatever one makes of the arguments here, the task of working out what these systems reveal about the mental, and what they do not, is one that will occupy philosophy for years to come.

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